

Homework 5: Chi-Square Tests and ANOVAs

Note: These are tab-separated files. See [here](#) for how to use the read_tsv function. Datasets are linked at the beginning of each question. Don't forget to test your assumptions!

R Code

Download R Code

1. [\(zodiac.txt\) Fortune magazine collected the zodiac signs](#)
2. [Download \(zodiac.txt\) Fortune magazine collected the zodiac signs](#)
3. of 256 heads of the largest 400 companies.
The data shows the number of births for each sign. Choose the appropriate Chi-square test to determine if some zodiac signs are over-represented in the heads of large companies. [Helpful video for running the test](#)
4. [\(analgesics.txt\) A pharmaceutical company tested three formulations](#)
5. [Download \(analgesics.txt\) A pharmaceutical company tested three formulations](#)
6. of a pain relief medicine for migraine headache sufferers. For the experiment, 27 volunteers were selected and 9 were randomly assigned to one of three drug formulations. The subjects were instructed to take the drug during their next migraine headache episode and to report their pain on a scale of 1 = no pain to 10 = extreme pain 30 minutes after taking the drug. Perform a one-way ANOVA to determine if any of the medications performed better than the others. If yes, perform a Tukey's HSD to determine which was/were significantly different.
7. [\(tv_watching.csv\) A study at a liberal arts college](#)
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9. attempted to find out whether men and women watch the same amount of TV, on average and whether it mattered if students were varsity athletes or not. Student researchers asked 200 randomly selected students questions about their backgrounds and about their television-viewing habits and received 197 legitimate responses. The researchers found that men watch, on average, about 2.5 hours per week more TV than women, and that varsity athletes watch about 3.5 hours per week more than those who are not varsity athletes. But is this the whole story? To investigate further, they divided the students into four groups: male athletes (MA), male non-athletes (MNA), female athletes (FA), and female non-athletes (FNA). Perform an additive two-way ANOVA and a two-way ANOVA with interaction, examine model fit to determine if one model performed better than the other. Generate a summary

of the best performing model and if it shows statistical significance, perform a Tukey's HSD to determine which factors was/were significantly different.